

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

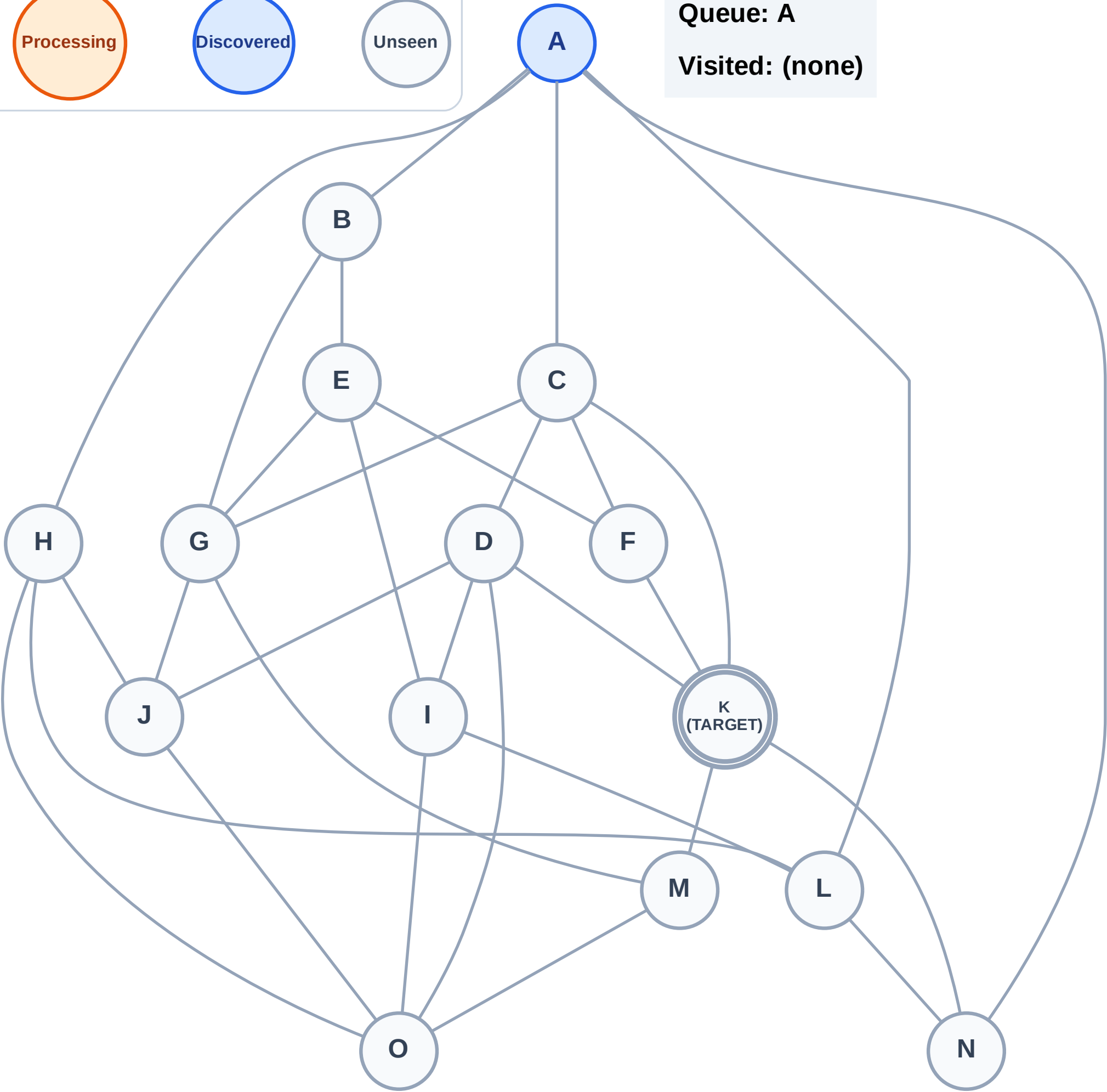
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

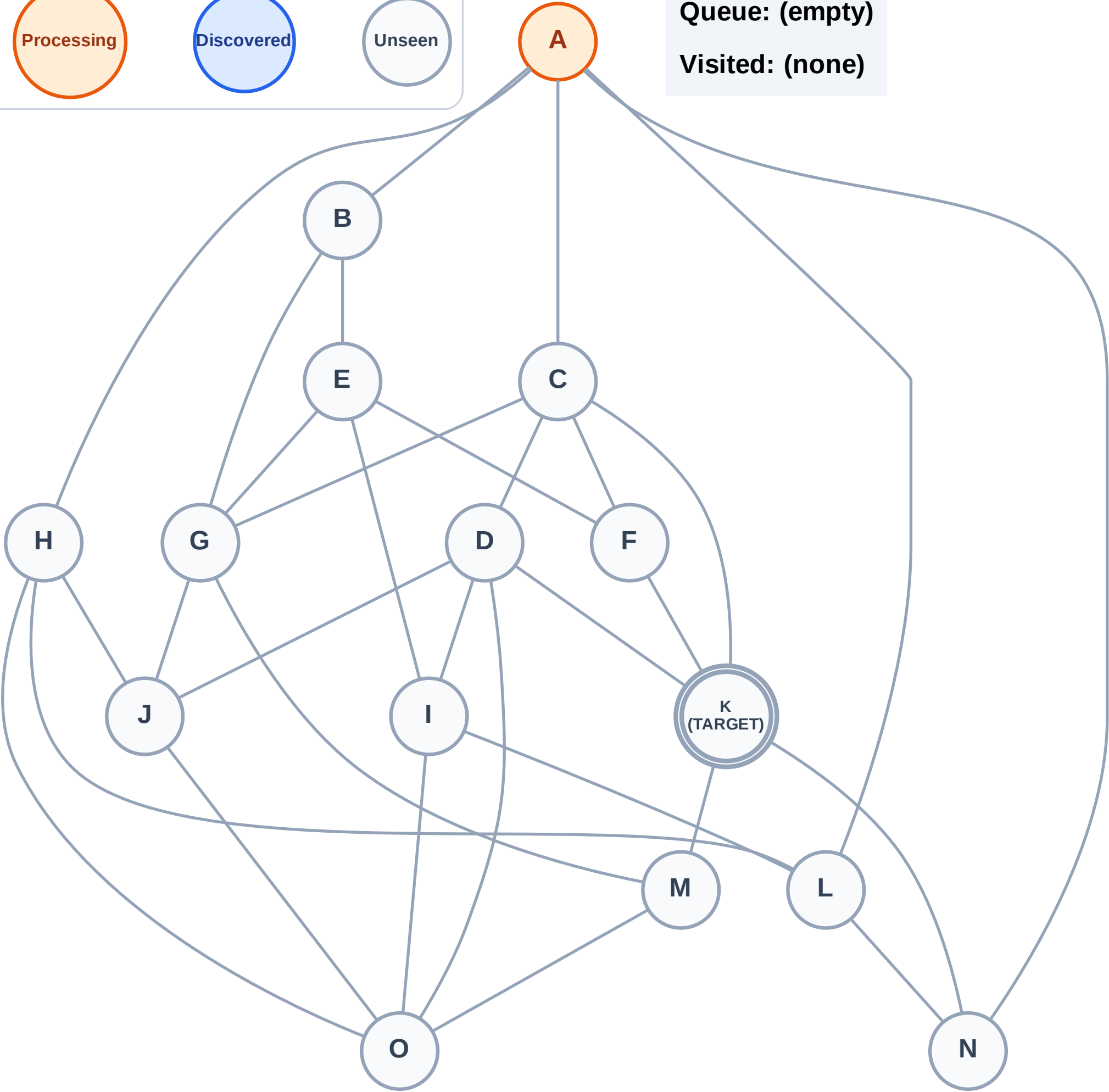
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

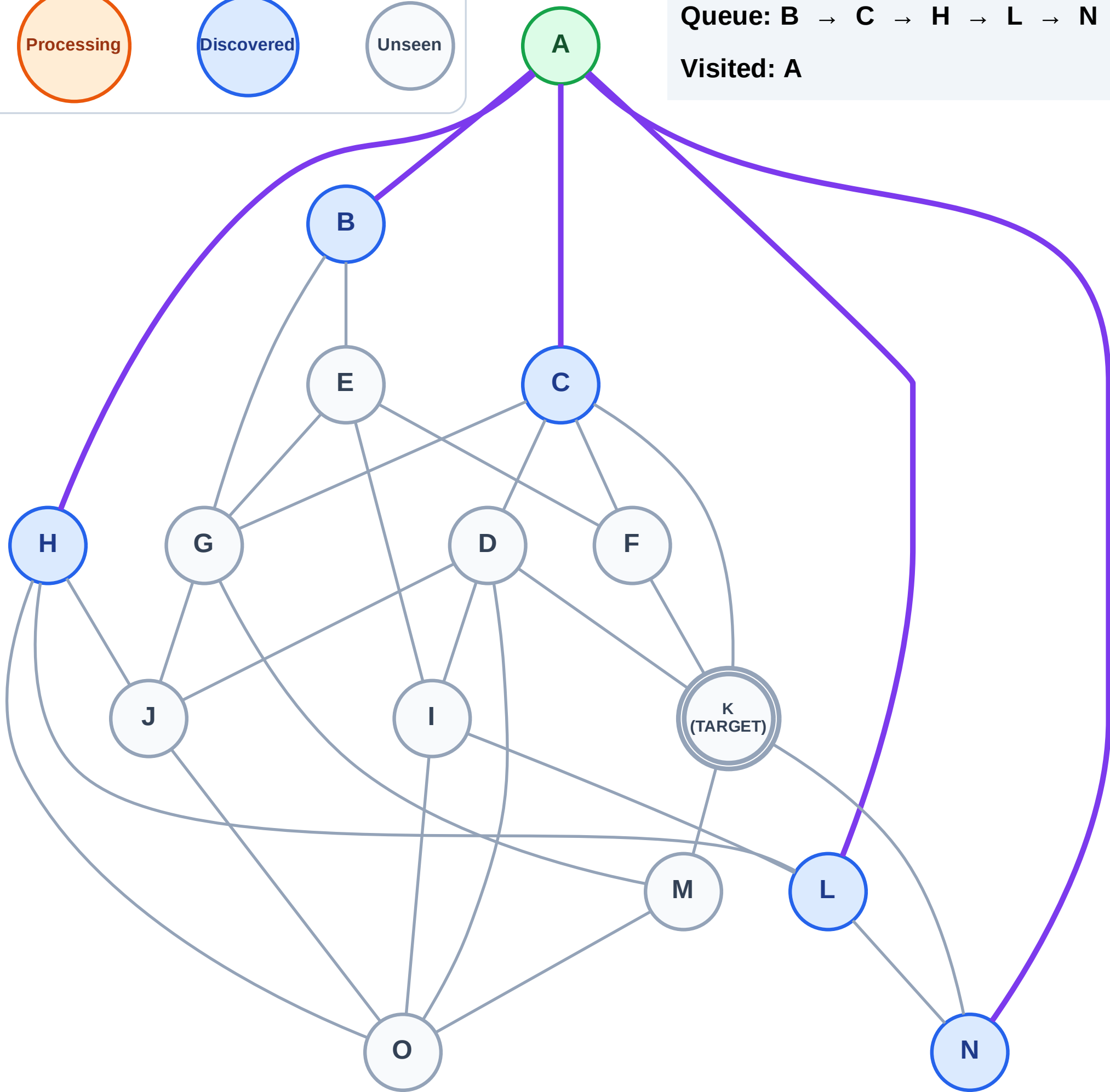
Step 3: Discover B, C, H, L, N from A

Legend



Queue: B → C → H → L → N

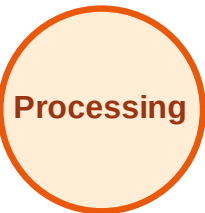
Visited: A



BFS Traversal

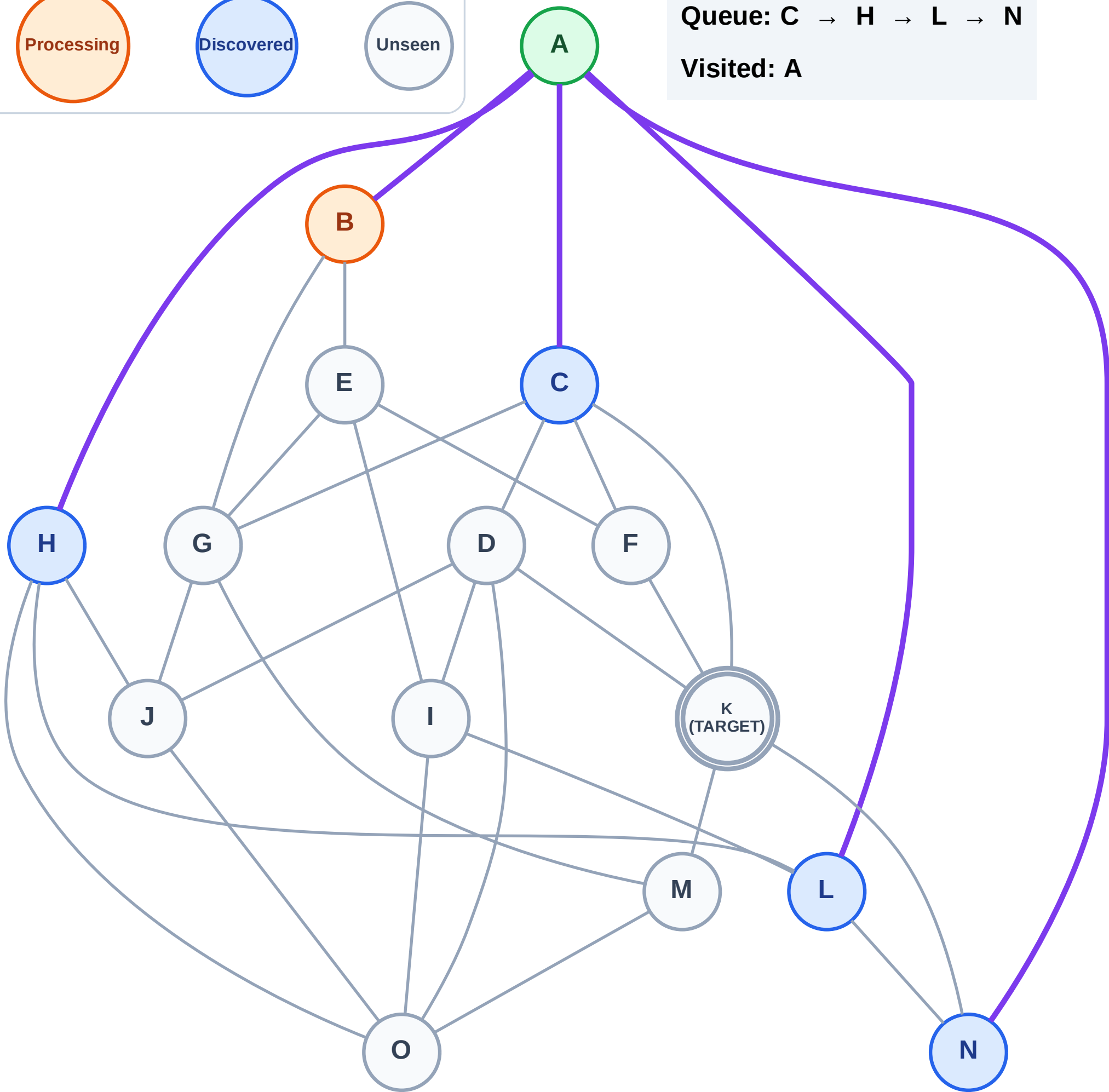
Step 4: Process node B

Legend



Queue: C → H → L → N

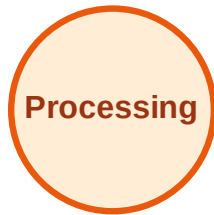
Visited: A



BFS Traversal

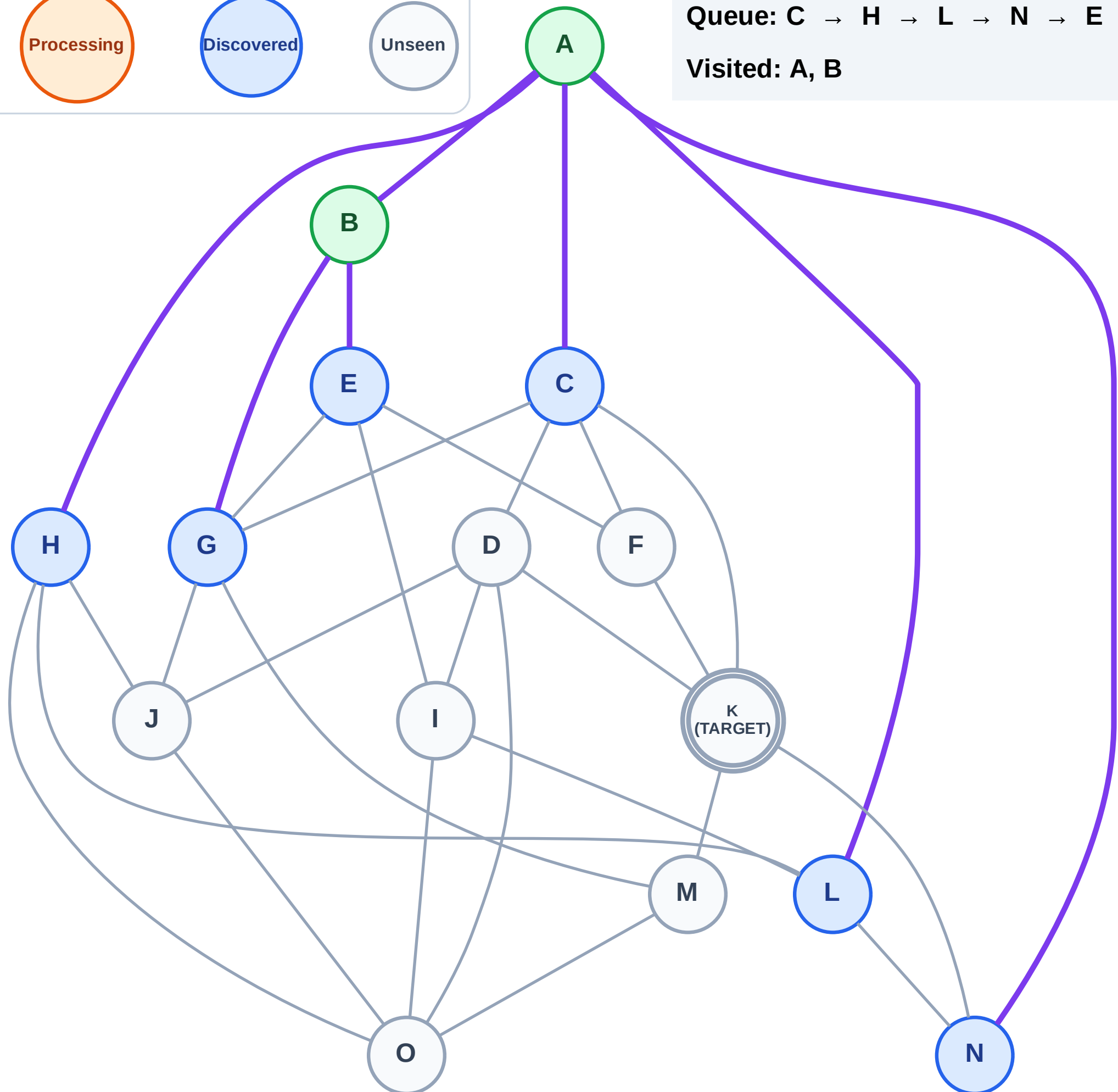
Step 5: Discover E, G from B

Legend



Queue: C → H → L → N → E → G

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

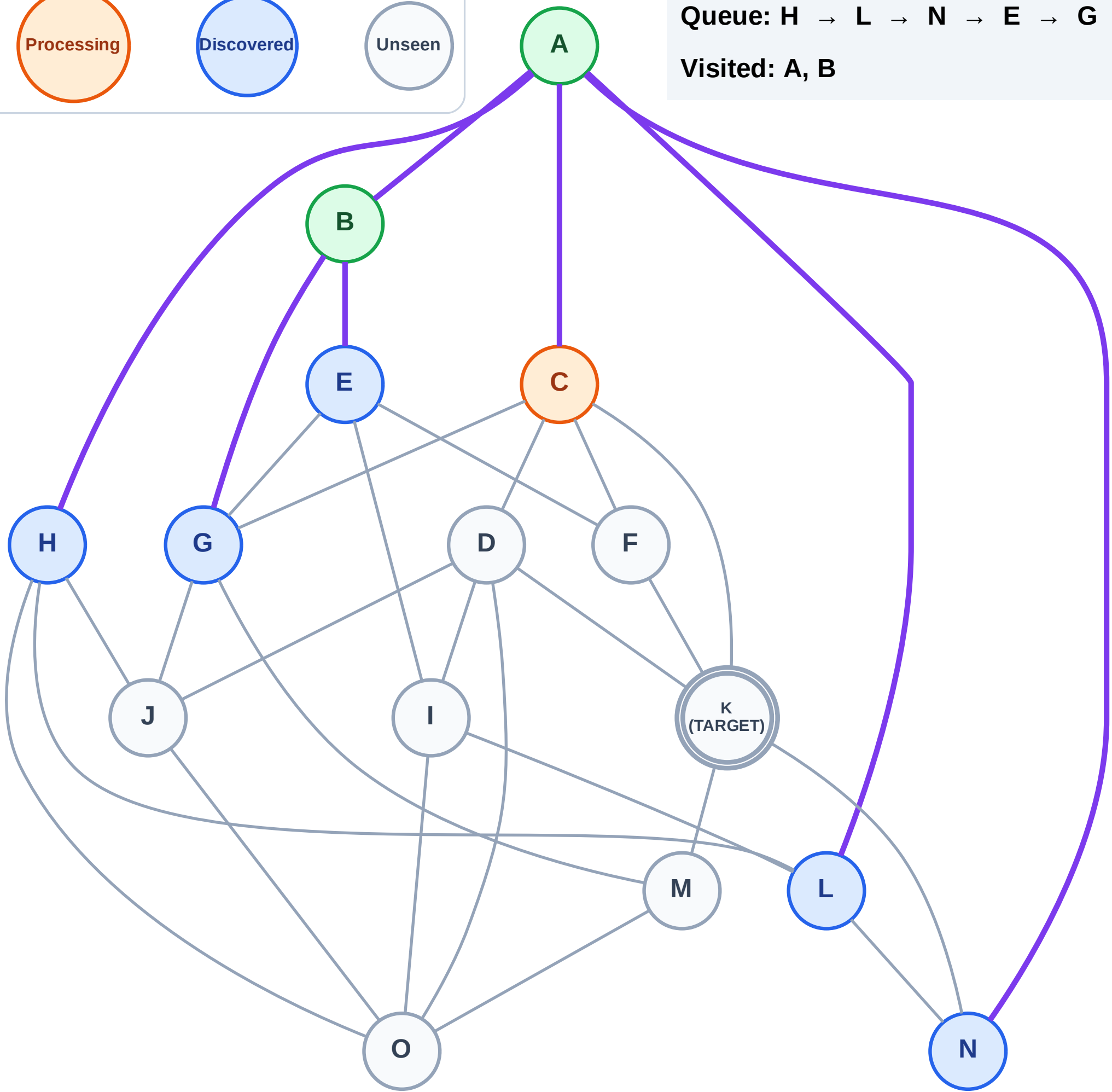
Processing

Discovered

Unseen

Queue: H → L → N → E → G

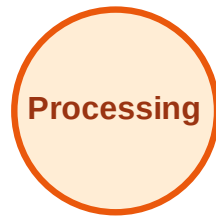
Visited: A, B



BFS Traversal

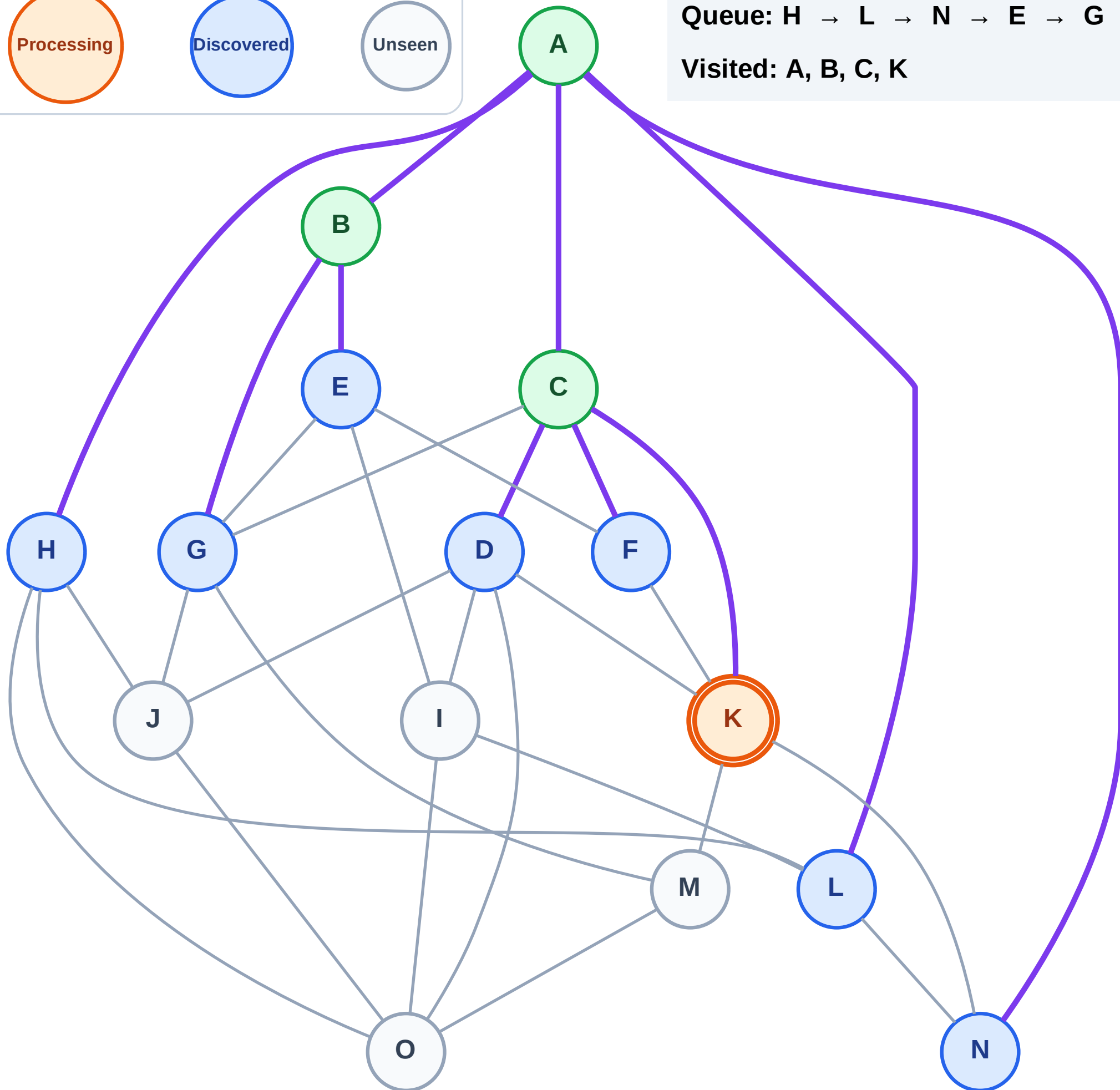
Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend



Queue: H → L → N → E → G → D → F

Visited: A, B, C, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

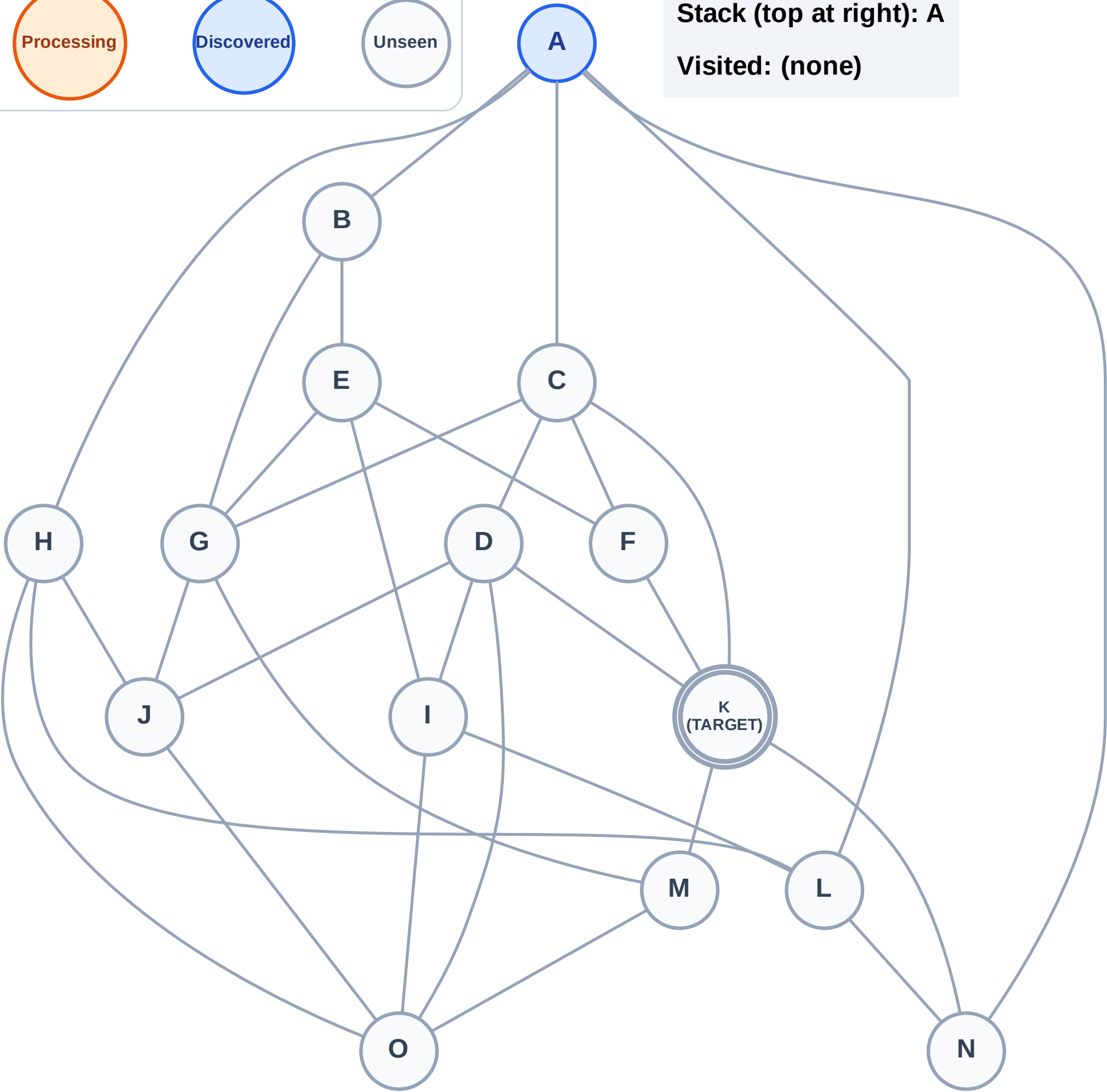
Complete

Processing

Discovered

Unseen

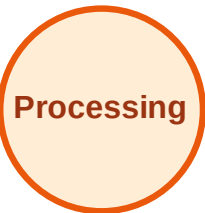
Stack (top at right): A
Visited: (none)



DFS Traversal

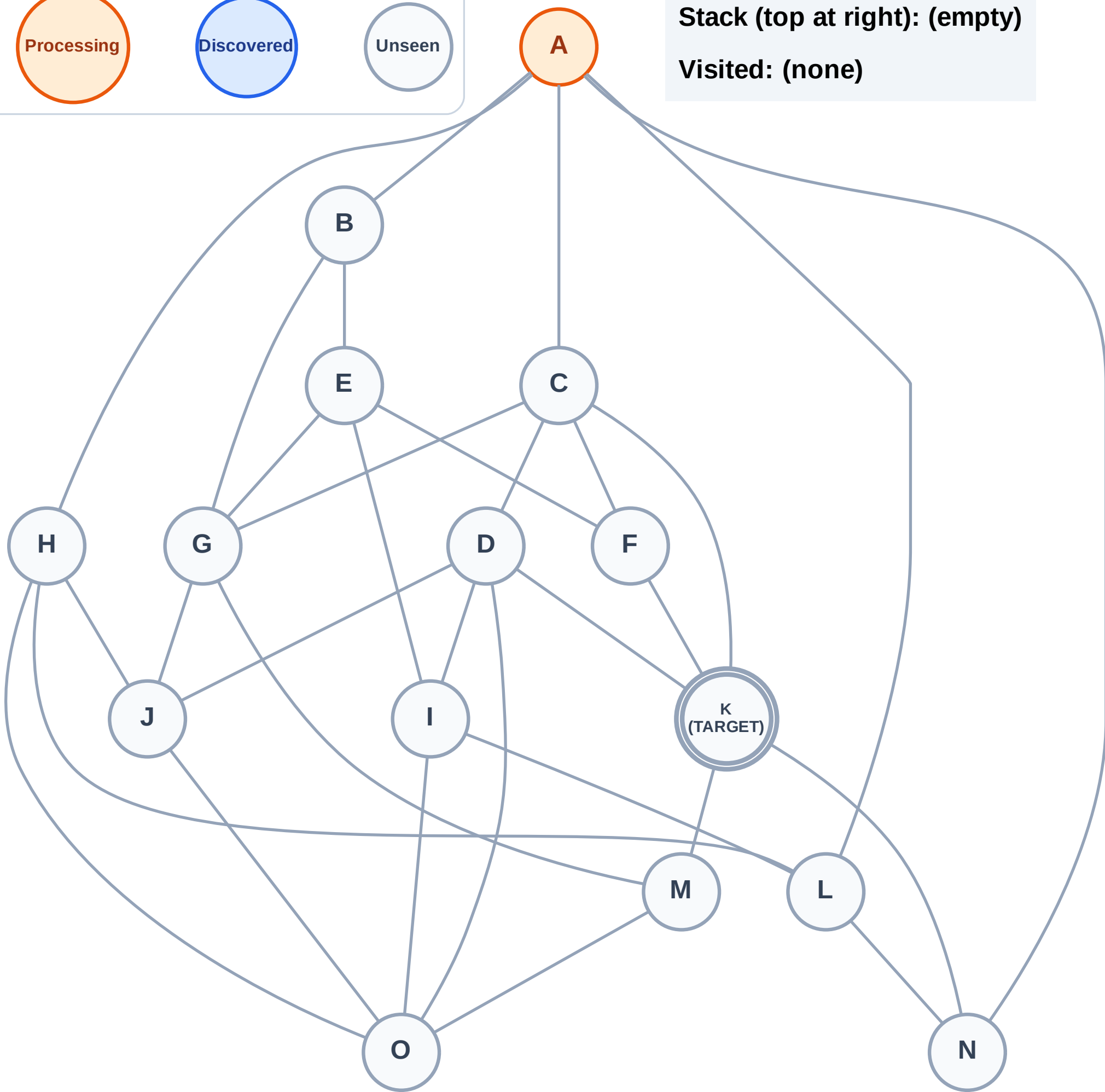
Step 2: Process node A

Legend



Stack (top at right): (empty)

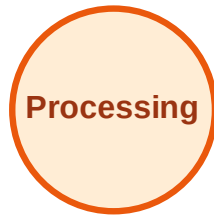
Visited: (none)



DFS Traversal

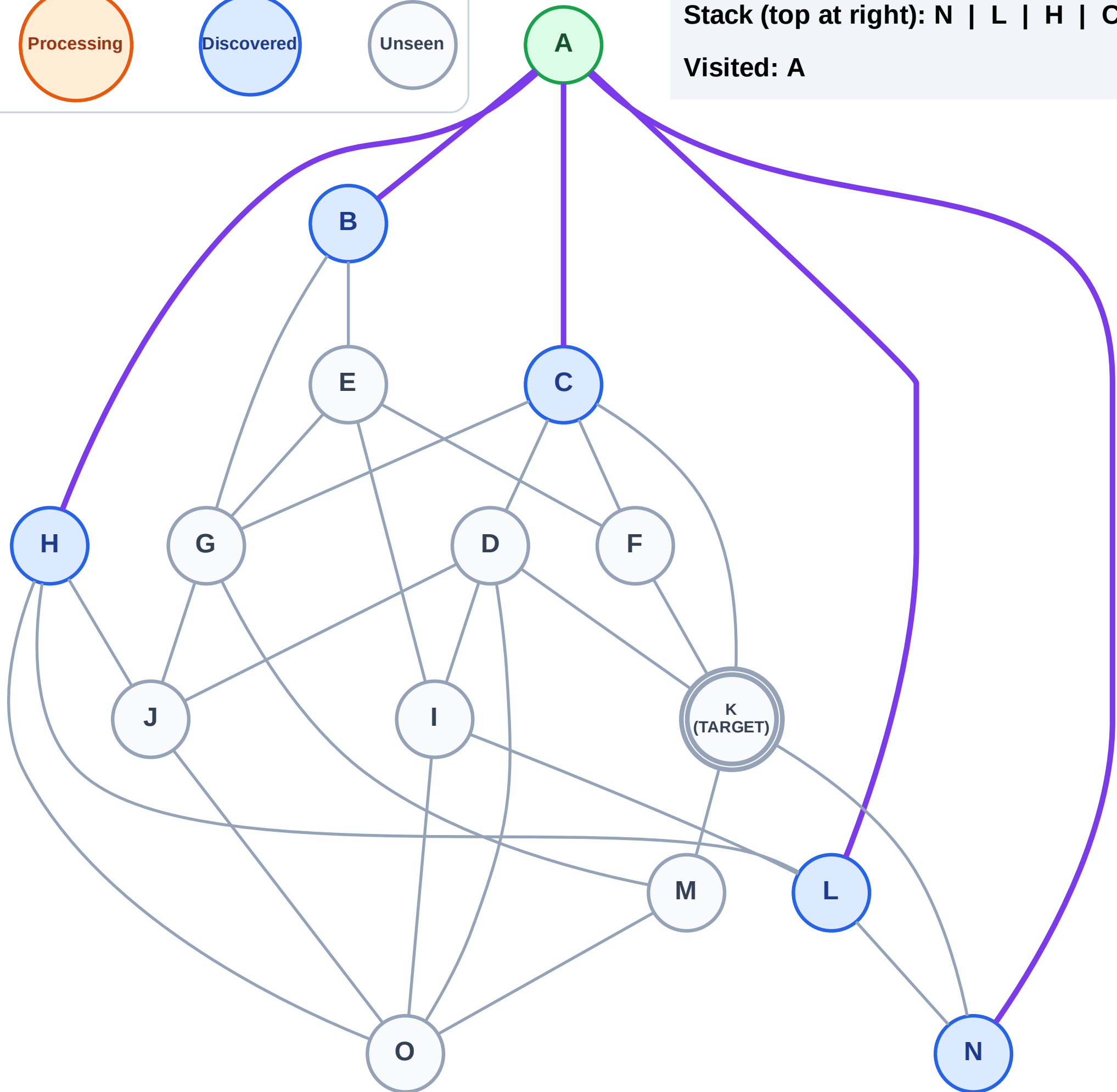
Step 3: Discover N, L, H, C, B from A

Legend



Stack (top at right): N | L | H | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

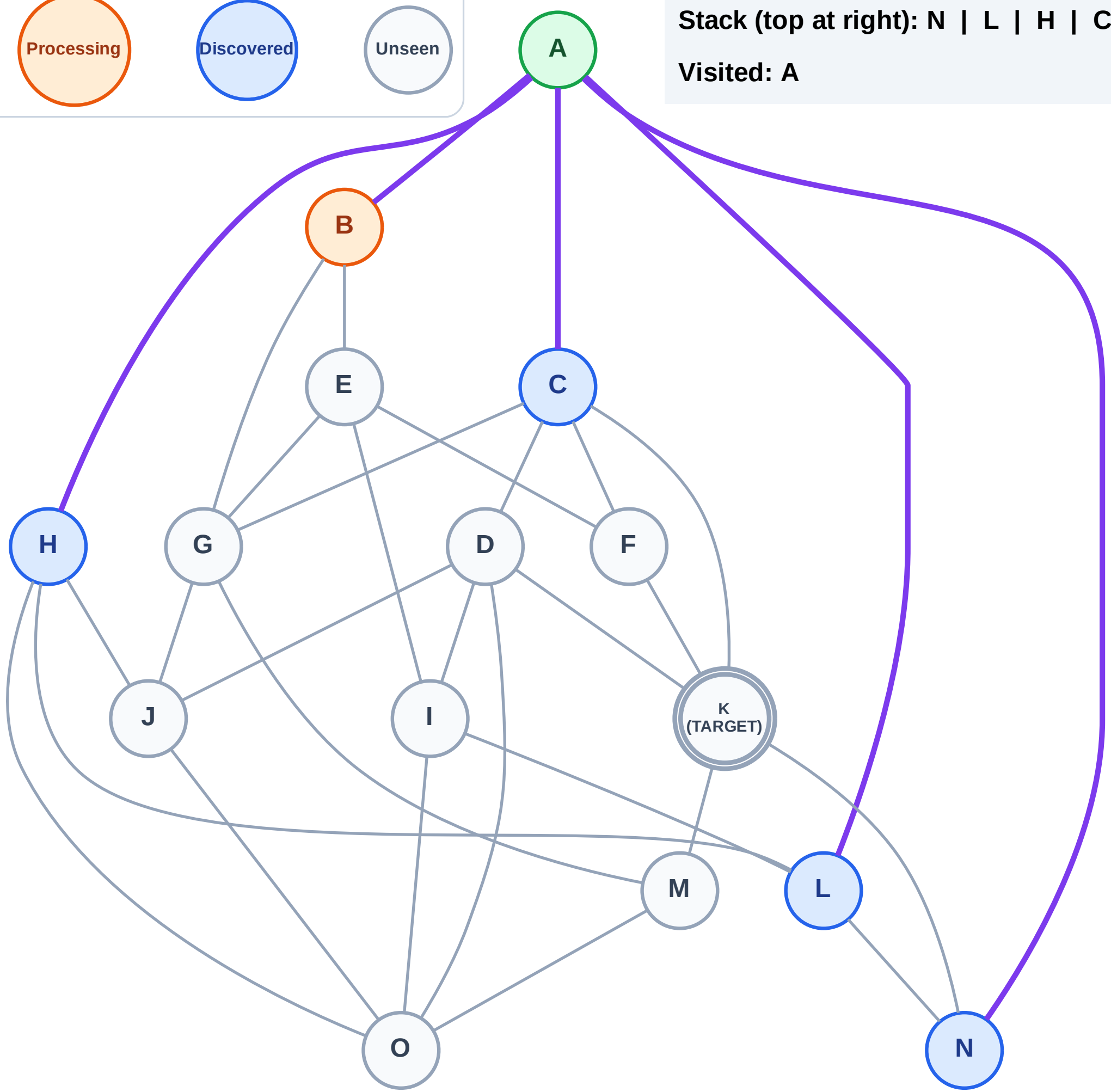
Complete

Processing

Discovered

Unseen

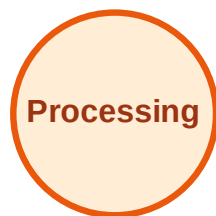
Stack (top at right): N | L | H | C
Visited: A



DFS Traversal

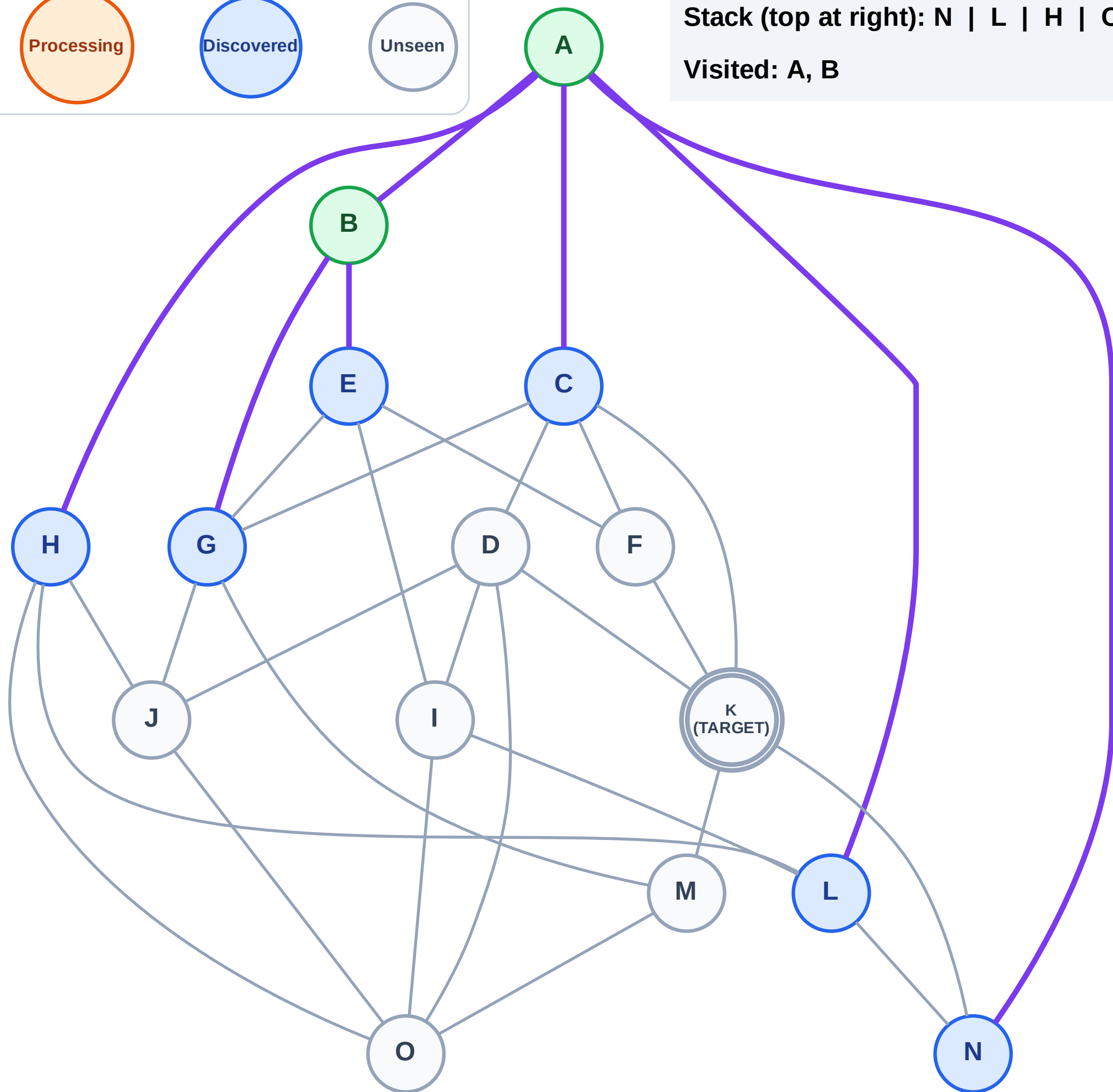
Step 5: Discover G, E from B

Legend



Stack (top at right): N | L | H | C | G | E

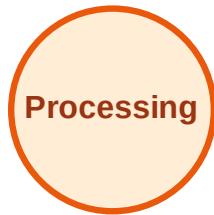
Visited: A, B



DFS Traversal

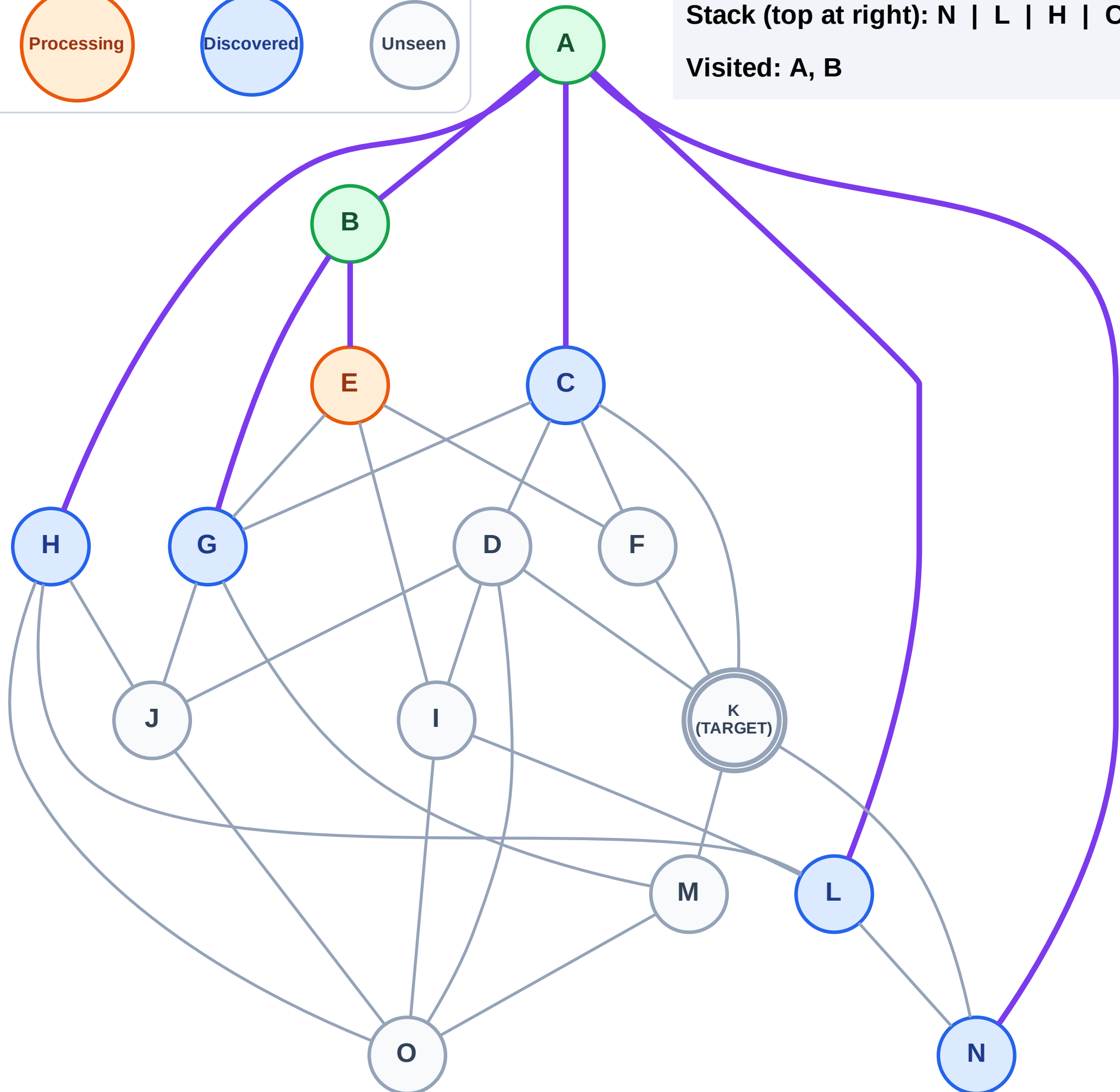
Step 6: Process node E

Legend



Stack (top at right): N | L | H | C | G

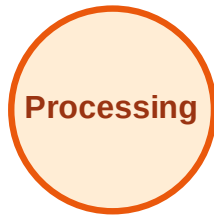
Visited: A, B



DFS Traversal

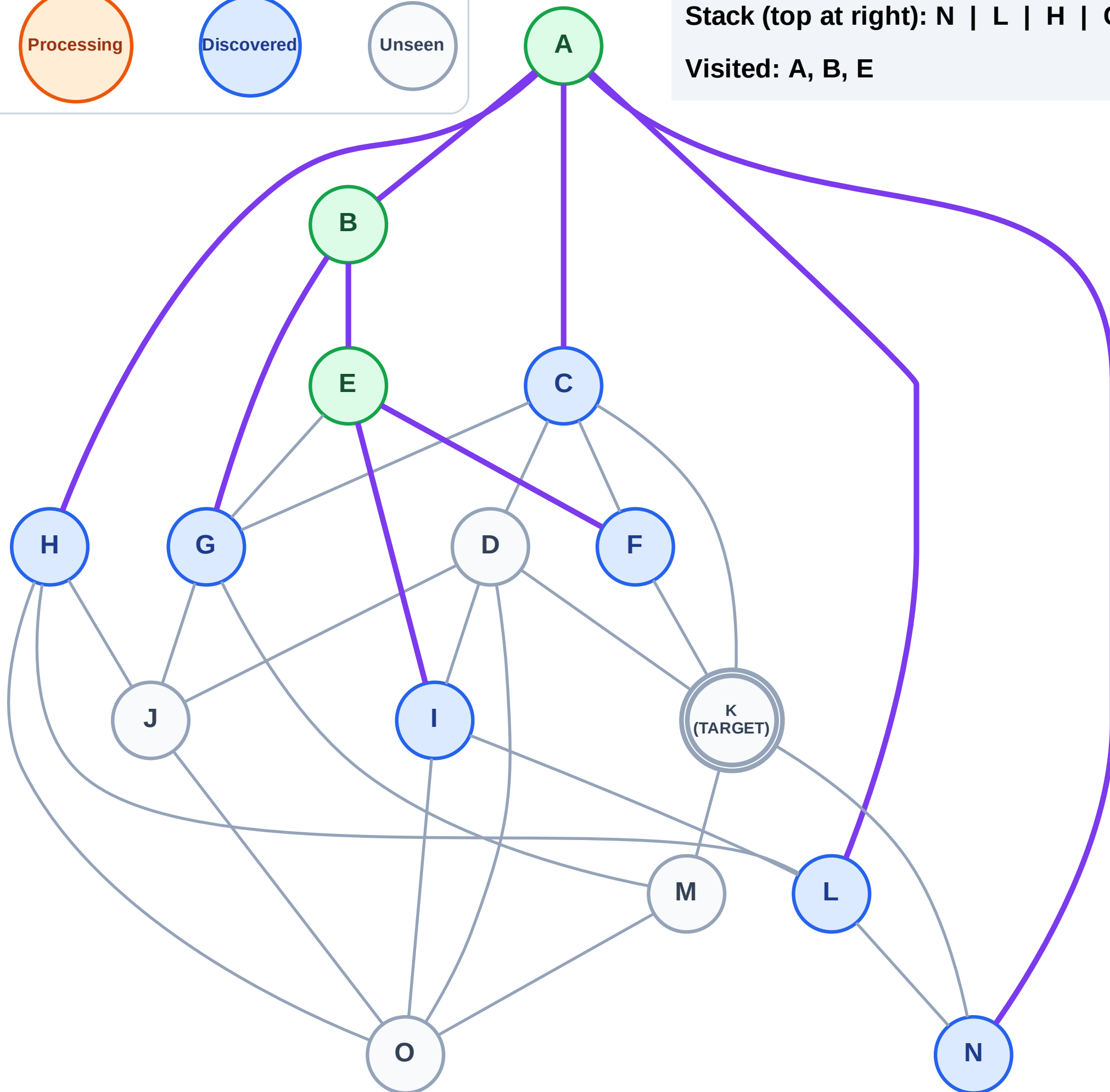
Step 7: Discover I, F from E

Legend



Stack (top at right): N | L | H | C | G | I | F

Visited: A, B, E



DFS Traversal

Step 8: Process node F

Legend

Complete

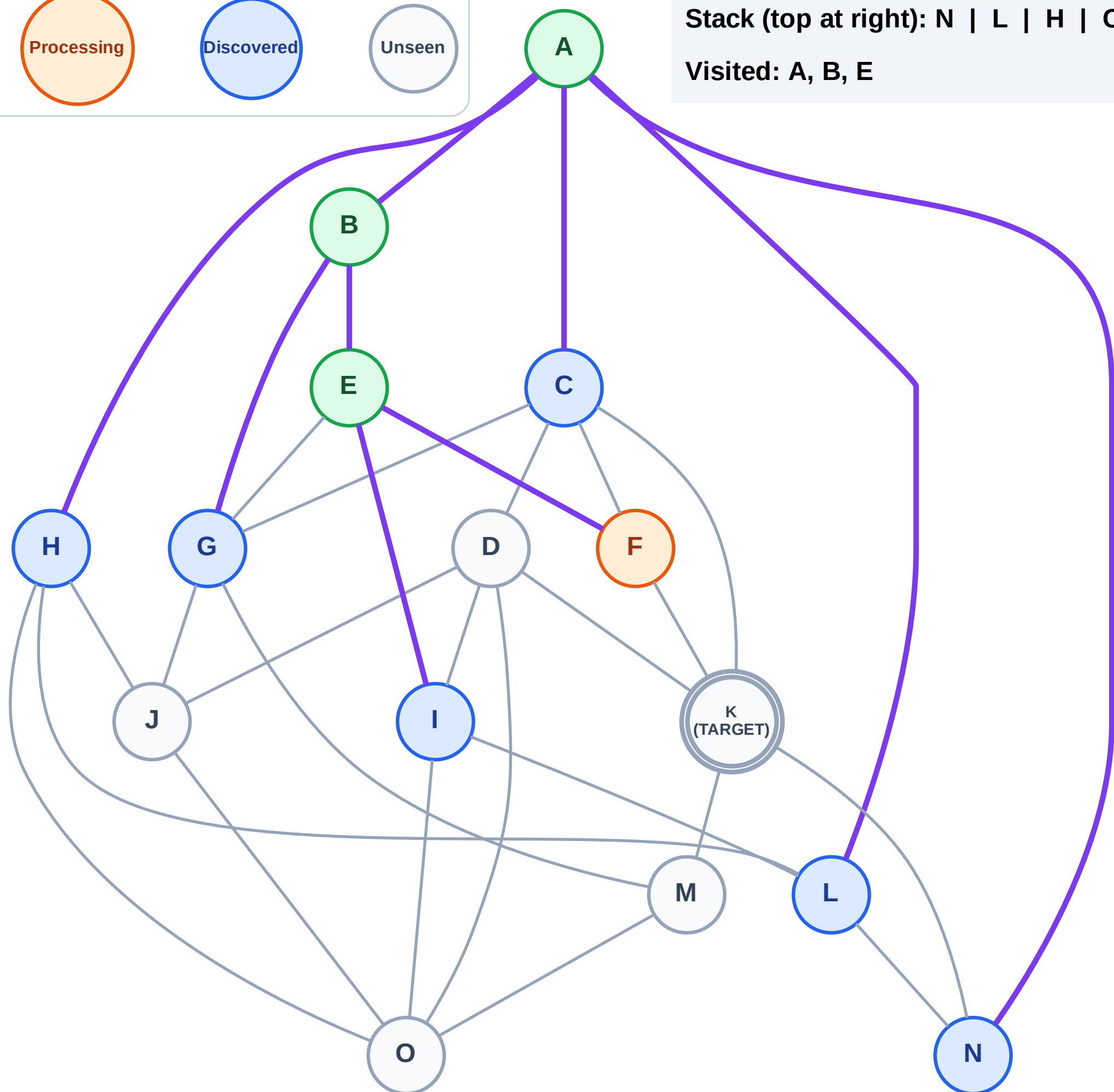
Processing

Discovered

Unseen

Stack (top at right): N | L | H | C | G | I

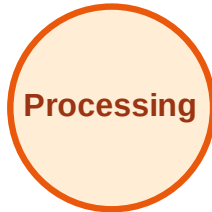
Visited: A, B, E



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → E → F → K

Legend



Stack (top at right): N | L | H | C | G | I

Visited: A, B, E, F, K

